The coefficient relations for squared Berndt integrals MathDB 357280

Research note

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Task 1. Prove both coefficient relations

Put $G = \Gamma(1/4)$. For every integer $p \geq 1$, the two evaluations in Xu–Zhao [1, Theorems 6.3 and 8.12] have the form

$$I_p^- := \int_0^\infty \frac{x^{4p+1}\,dx}{(\cos x - \cosh x)^2} = i_p \frac{G^{8p}}{\pi^{2p}} + j_p \frac{G^{8p+8}}{\pi^{2p+4}}, \tag{1}$$

$$I_p^+ := \int_0^\infty \frac{x^{4p+1}\,dx}{(\cos x + \cosh x)^2} = g_p \frac{G^{8p}}{\pi^{2p}} + h_p \frac{G^{8p+8}}{\pi^{2p+4}}, \tag{2}$$

with rational coefficients. This specifies the theorem references omitted from the MathDB transcription. We prove Conjecture 8.14 of the current source, without changing its range or conclusion.

Theorem 1. *For every positive integer p,*

$$\boxed{g_p = -\frac{2^{2p-1} - (-1)^p}{2^{2p-1}}\, i_p, \qquad h_p = -\frac{2^{2p-1} + (-1)^p}{2^{2p-1}}\, j_p.}$$

The proof also constructs all four rational coefficients. The essential connection between the two integral families is the degree-two Eisenstein identity, followed by a modular transformation from i to $(1+i)/2$.

1. A common residue calculation

Fix $p \geq 1$ and write $w = 4p$, $a = (-1)^p 2^{2p-1}$. For $y > 0$ set

$$S_-(y) = \sum_{n \geq 1} n^w \operatorname{csch}^2(ny), \qquad S_+(y) = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} k^w \operatorname{sech}^2(ky/2).$$

We first establish

$$I_p^- = a\pi^{w+1}\{(w+1)S_-(\pi) + \pi S_-'(\pi)\}, \tag{3}$$

$$I_p^+ = -\frac{a}{2^w}\pi^{w+1}\{(w+1)S_+(\pi) + \pi S_+'(\pi)\}. \tag{4}$$

These formulas can also be read from the source's contour results; we give the details to fix their relative sign and scale.

Let $a_0 = (1+i)/2$, $b_0 = (1-i)/2$. The factorizations are

$$\cos z + \cosh z = 2\cosh(a_0 z)\cosh(b_0 z), \qquad \cos z - \cosh z = -2\sinh(a_0 z)\sinh(b_0 z).$$

Use positively oriented triangles with vertices $0, R, iR$, taking $R = 2\pi N$ for the plus sign and $R = (2N+1)\pi$ for the minus sign. The first hyperbolic factor on the oblique edge has modulus at least one, and the second has modulus at least $\sinh(R/2)$. The oblique integral is $O_p(R^{w+2}e^{-R})$ and tends to zero. The two axial integrals have the same sign and sum to twice the real integral. For the minus sign the apparent singularity at zero is removable: the quotient is $O(z^{4p-3})$ there. Thus for either sign the integral equals πi times the sum of its first-quadrant residues.

For the minus sign the poles are $(1+i)t$, $t = n\pi$, $n \geq 1$. Writing $z = (1+i)t + v$, the denominator squared is $4\sinh^2(a_0 v)\sinh^2(t + b_0 v)$. Since $b_0(1+i) = 1$ and $(1+i)^w = (-4)^p$, its residue is

$$-ia\{(w+1)t^w \operatorname{csch}^2 t + t^{w+1}(\operatorname{csch}^2)'(t)\}.$$

For the plus sign $t = (n+1/2)\pi$, $n \geq 0$, the local denominator squared is $-4\sinh^2(a_0 v)\cosh^2(t + b_0 v)$, and the residue is

$$ia\{(w+1)t^w \operatorname{sech}^2 t + t^{w+1}(\operatorname{sech}^2)'(t)\}.$$

Summing and multiplying by πi proves (3)–(4). All sums and their differentiated versions converge locally uniformly for $y > 0$ by exponential decay.

2. The degree-two Eisenstein identity

Use $q = e^{2\pi i\tau}$ and $D = (2\pi i)^{-1} d/d\tau = q\, d/dq$. Let

$$f(\tau) = E_w(\tau) = 1 - \frac{2w}{B_w}\sum_{n\geq 1}\sigma_{w-1}(n)q^n, \qquad c = -\frac{B_w}{2w}.$$

Then $L_{w-1}(q) = \sum_{n\geq 1} n^{w-1}q^n/(1-q^n) = c(f-1)$. For $\tau = iy/\pi$, expansion of csch^2 immediately gives

$$S_-(y) = 4c(Df)(\tau). \tag{5}$$

Selecting odd k in the expansion of sech^2 similarly gives

$$S_+(y) = 4c\{(Df)(\tau/2) - (2^w+4)(Df)(\tau) + 2^{w+2}(Df)(2\tau)\}. \tag{6}$$

For example, the selector identity before differentiation is

$$\sum_{k \text{ odd}} \frac{k^{w-1}q^k}{1+q^k} = L_{w-1}(q) - (2+2^{w-1})L_{w-1}(q^2) + 2^w L_{w-1}(q^4).$$

Its derivative gives (6), including the chain factors.

The elementary divisor identity, with $\sigma_r(n/2) = 0$ for odd n,

$$\sigma_r(2n) = (1+2^r)\sigma_r(n) - 2^r\sigma_r(n/2)$$

proves coefficient by coefficient that

$$f(\tau/2) + f((\tau+1)/2) + 2^w f(2\tau) = (2+2^w)f(\tau). \tag{7}$$

Indeed the two half-argument terms select the even Fourier indices; the constant terms also agree. Differentiate (7) and substitute in (6) to obtain the key simplification

$$S_+(y) = 4c\{2^w(Df)(\tau) - (Df)((\tau+1)/2)\}. \tag{8}$$

3. Two transformed derivatives

Write $f_r = (D^r f)(i)$ for $r = 0, 1, 2$, and put

$$K = (-1)^p 2^{2p} = (1+i)^w, \qquad t = \frac{2K}{2^w} = \frac{(-1)^p}{2^{2p-1}}.$$

The transformation $\gamma\tau = \tau/(\tau+1)$ is in $\mathrm{SL}_2(\mathbb{Z})$ and sends i to $(1+i)/2$. For any modular form f of weight w, differentiating $f(\gamma\tau) = (\tau+1)^w f(\tau)$ once and twice gives, with $\lambda = (2\pi i)^{-1}$,

$$\begin{aligned}(Df)(\gamma\tau) &= (\tau+1)^{w+2} Df(\tau) + w\lambda(\tau+1)^{w+1} f(\tau),\\ (D^2 f)(\gamma\tau) &= (\tau+1)^{w+4} D^2 f(\tau) + 2(w+1)\lambda(\tau+1)^{w+3} Df(\tau)\\ &\quad + w(w+1)\lambda^2(\tau+1)^{w+2} f(\tau).\end{aligned}$$

Differentiating $f(-1/\tau) = \tau^w f(\tau)$ at $\tau = i$ gives $f_1 = w f_0/(4\pi)$. Substituting this and $(1+i)^2 = 2i$ above yields

$$(Df)((1+i)/2) = 2Kf_1, \qquad (D^2 f)((1+i)/2) = K\left(-4f_2 + \frac{w(w+1)}{2\pi^2} f_0\right). \tag{9}$$

Now $d/dy = -2D$ for the argument $\tau = iy/\pi$, whereas it is $-D$ for $(\tau+1)/2$. Equations (5), (8), and (9) give

$$\begin{aligned} S_-(\pi) &= 4cf_1, & S'_-(\pi) &= -8cf_2,\\ S_+(\pi) &= 4c(2^w - 2K)f_1, & S'_+(\pi) &= -8c\left((2^w+2K)f_2 - \frac{Kw(w+1)}{4\pi^2} f_0\right).\end{aligned}$$

Using $f_1 = w f_0/(4\pi)$ in the residue formulas, the extra Kf_0 terms cancel. We obtain

$$I_p^- = 4ca\pi^{w+1}\{(w+1)f_1 - 2\pi f_2\}, \tag{10}$$

$$I_p^+ = -4ca\pi^{w+1}\{(w+1)f_1 - 2\pi(1+t)f_2\}. \tag{11}$$

4. Separate the two period powers

Set $u = G^4/(4\pi^3)$. The classical values are

$$E_2(i) = 3/\pi, \qquad E_4(i) = 3u^2/4, \qquad E_6(i) = 0.$$

The rational modular-form ring is $\mathbb{Q}[E_4, E_6]$ [2, Proposition 4]. Thus $f_0 = \alpha u^{2p}$ for some $\alpha \in \mathbb{Q}$. For a weight-k modular form F, its Serre derivative

$$\vartheta_k F = DF - \frac{k}{12} E_2 F$$

is modular of weight $k+2$, with rational Fourier coefficients when F has them [2, Section 5.1]. Put $F = \vartheta_w f$ and $H = \vartheta_{w+2} F$. Since a weight congruent to two modulo four vanishes at i, $F(i) = 0$. Ramanujan's identity $DE_2 = (E_2^2 - E_4)/12$ therefore gives

$$\begin{aligned} f_2 &= \frac{w}{12}(DE_2)(i) f_0 + \frac{w}{12} E_2(i) f_1 + H(i)\\ &= \frac{w(w+1)}{16\pi^2} f_0 + H(i) - \frac{w}{144} E_4(i) f_0.\end{aligned}$$

The last two terms are the value at i of a rational modular form of weight $w+4$, so they equal βu^{2p+2} for some $\beta \in \mathbb{Q}$. Hence

$$f_0 = \alpha u^{2p}, \qquad f_1 = \frac{w}{4\pi} f_0, \qquad f_2 = \frac{w(w+1)}{16\pi^2} f_0 + \beta u^{2p+2}. \tag{12}$$

Put $L = w(w+1)f_0/(8\pi)$ and $U = -2\pi\beta u^{2p+2}$. Substitution of (12) in (10)–(11) gives

$$I_p^- = 4ca\pi^{w+1}(L+U), \qquad I_p^+ = -4ca\pi^{w+1}\{(1-t)L + (1+t)U\}.$$

The first part is a rational multiple of G^{8p}/π^{2p}, and the second is a rational multiple of G^{8p+8}/π^{2p+4}. These two numbers are linearly independent over $\mathbb{Q}$, by the established algebraic independence of π and $\Gamma(1/4)$ [2, p. 85]. Comparison with (1)–(2) therefore proves $g_p = -(1-t)i_p$ and $h_p = -(1+t)j_p$, as required. All steps apply to every $p \geq 1$. □

For an explicit coefficient algorithm, the same calculation gives

$$i_p = \frac{(-1)^p cw(w+1)\alpha}{2^{2p+2}}, \qquad j_p = \frac{(-1)^{p+1} c\beta}{2^{2p+2}}.$$

The quantities α, β follow by rational polynomial arithmetic in E_4, E_6 and the three Ramanujan differential equations. For example, $p = 1$ gives $c = 1/240$, $\alpha = 3/4$, $\beta = 5/64$, and

$$(i_1, j_1, g_1, h_1) = \left(-\frac{1}{256}, \frac{1}{3\cdot 2^{14}}, \frac{3}{512}, -\frac{1}{3\cdot 2^{15}}\right),$$

in agreement with the source examples. This checks the smallest allowed index without relying on an asymptotic or generic-index argument.

Task 2. Generalize the modular correspondence

Replace the degree-two selector identity by the degree-N Eisenstein identity. Identify integral families whose residue sums realize the resulting transformed derivatives, and derive their coefficient relations.

Task 3. Compare higher denominator powers

Find the analogue of the two coefficient relations when the denominator power is four or higher. Determine which finite combinations of modular derivatives control each period power, including contributions from the additional poles and the convergence condition at zero.

Task 4. Determine coefficient signs and growth

Determine the signs and possible vanishing of β in (12) for all p. Obtain asymptotic formulas for i_p, j_p with explicit error bounds, and use the proved relations to transfer them to g_p, h_p.

Task 5. Derive recurrences for exact coefficients

Construct recurrences for α and β as the weight increases by four. Seek a method computing the four integral coefficients without reconstructing the full Eisenstein polynomial at each new weight.

References

[1] C. Xu and J. Zhao, *Berndt-Type Integrals and Series Associated with Ramanujan and Jacobi Elliptic Functions*, arXiv:2301.08211v2 (2024), Theorems 6.3 and 8.12, Conjecture 8.14.

[2] D. Zagier, *Elliptic Modular Forms and Their Applications*, in *The 1-2-3 of Modular Forms*, Springer (2008), pp. 1–103; author-hosted chapter.

Verification and status. Run `verify_relations.py` for exact symbolic checks and separate numerical diagnostics. The proof for all p is the analytic and modular argument above, not a finite test. Full current-source and dated bounded claim checks accompany this note. The authoring agent conducted the proof audit; independent peer review has not occurred.